

【9.25 | Economist】 Actions and words **动作与言语**

难度评级: ★★★☆☆

【考研那些事儿之科技系列】

今日文章脉络:

脑机接口技术发展迅速 (Para. 1)



新研究已实现瘫痪者通过虚拟形象交流 (Para. 2)



研究方案及存在的问题 (Para. 3)



该研究首次实现言语和手势同步解码,
使患者的交流更富有表现力 (Para. 4)



研究尚待推进, 解码更多动作,
真正改变瘫痪者的生活 (Para. 5)

A new device helps severely **paralysed** patients communicate

新型设备助力严重瘫痪患者交流

【学习流程：第一遍请先看原文自己精读，第二遍对照着本精读文档进行理解。文章标注的单词为重点单词，只提供相关注释，想要详细释义麻烦同学查一下词典哦！另外文章底色为灰色的句子为今天的翻译作业，今日文档中的句子分析模块会提供答案，同学们请先思考再看句子分析~】

【Para.1】

Brain-computer **interfaces** have **come a long way** quickly. The first successful **decoding** of **vowel** sounds from a neural **implant** was reported in 2009. In 2021 a team from Edward Chang's lab at the University of California, San Francisco decoded the words and sentences of a patient who could not speak.

- **paralyse v.** /'pærəlaɪz/ to make sb unable to feel or move all or part of their body 使瘫痪；使麻痹 【考研大纲词汇】

The accident left him paralysed from the waist down. 那场事故使他腰部以下都瘫痪了。

- **interface n.** an electrical circuit , connection or program that joins one device or system to another 接口；接口程序；连接电路 【考研大纲词汇】

the interface between computer and printer 计算机和打印机之间的接口

- **come a long way phrase** 有了很大进步或发展

We've come a long way since the early days of the project.

这项目开始以来我们已取得很大进展。

- **decode v.** if you decode a message that has been written or spoken in a code, you change it into ordinary language.

解码

All he had to do was decode it and pass it over. 他所要做的就是解译它，然后把它发过去。

- **vowel n.** /'vaʊəl/ a letter that represents a vowel sound. In English the vowels are a, e, i, o, and u 元音字母

- **implant n.** /ɪm'plænt/ something that is put into a person's body in a medical operation (植入人体中的) 移植物, 植入物
silicone breast implants 硅酮乳房填充物

脑机接口技术发展迅速。2009 年，首次有研究报告成功从神经植入物中解码出元音字母。2021 年，加州大学旧金山分校爱德华·常 (Edward Chang) 实验室的一支团队解码了一名无法说话患者的单词和句子。

【Para.2】

A new paper by Dr Chang's group, published in Nature Neuroscience, describes the **passing** of another milestone. He and his colleagues have enabled two severely paralysed people to communicate with them, by means of an **avatar**, using both words and gestures. That is an **auspicious** result for the several million individuals around the world paralysed by **strokes**, **traumatic** injuries and **degenerative** illnesses such as motor-

neuron disease.

- **passing** n. /'pɑːsɪŋ/ **the ~ of time/the years** the process of time going by (时间、岁月的) 流逝, 推移 【考研大纲词汇】
I love him more with each passing day. 随着时间的流逝, 我越发爱他了。
- **avatar** n. /'ævətɑː(r)/ a picture of a person or an animal which represents a person, on a computer screen, especially in a computer game or chat room (尤指电脑游戏或聊天室中代表使用者的) 化身
- **auspicious** n. /ɔː'spiʃəs/ showing signs that sth is likely to be successful in the future 吉利的; 吉祥的
SYN promising
an auspicious start to the new school year 新学年的开门红
- **stroke** n. /strʊk/ a sudden serious illness when a blood vessel (= tube) in the brain bursts or is blocked, which can cause death or the loss of the ability to move or to speak clearly 中风 【考研大纲词汇】
to have/suffer a stroke 患中风
The stroke left him partly paralysed. 他因中风身体局部瘫痪了。
- **traumatic** adj. /trə'mætɪk/ (**psychology** 心 **medical** 医) connected with or caused by trauma 创伤的; 外伤的; 损伤的
traumatic amnesia 创伤性遗忘
- **degenerative** adj. (**of an illness** 疾病) getting or likely to get

worse as time passes (随着时间的推移) 变性的, 退化的
degenerative diseases such as arthritis 诸如关节炎之类的变
性病

常博士团队发表于《自然·神经科学》的新论文, 描述了又一个里程碑的实现。他和同事让两名严重瘫痪者能够借助虚拟形象, 运用言语和手势与研究人员交流。对于全球数百万因中风、创伤性损伤以及运动神经元病等退行性疾病而瘫痪的人而言, 这一成果令人鼓舞。

【Para.3】

The group did this by implanting, through a small **incision** in the skull of each patient, an array of sensors that covered a large part of the motor **cortex**—the brain region responsible for controlling **bodily** motion. This procedure took over two hours, in part because the process for connecting the sensor array to a data-carrying **cable** is **fiddly**. But the team are working on a **short-cut**, using a wireless device, that would reduce this to 40 minutes.

- **incision** n. /ɪnˈsɪʒ(ə)n/ a sharp cut made in sth, particularly during a medical operation; the act of making a cut in sth
切口; (尤指手术的) 切口; 切开
Make a small incision below the ribs. 在肋骨下方切开一个小口。
- **cortex** n. /ˈkɔːrteks/ (anatomy 解) the outer layer of an

organ in the body, especially the brain 皮层；皮质；（尤指）
大脑皮层

the cerebral/renal cortex 大脑/肾皮层

- **bodily adj.** connected with the human body 人体的；身体的
bodily functions/changes/needs 身体的机能/变化/需要
bodily fluids 体液

bodily harm (= physical injury) 对身体的伤害

- **cable n.** a set of wires, covered in plastic or rubber, that carries electricity, telephone signals, etc. 电缆
overhead/underground cables 高架/地下电缆

- **fiddly adj.** /'fɪdli/ difficult to do or use because small objects are involved 微小难弄的；需要手巧的；精巧难使用的
Changing a fuse is one of those fiddly jobs I hate. 换保险丝是我不愿干的麻烦事之一。

- **short-cut n.** a shortcut is a method of achieving something more quickly or more easily than if you use the usual methods 捷径

Fame can be a shortcut to love and money. 名望可以是得到爱情和金钱的捷径。

该团队采用的方法如下：经每名患者颅骨上的一个小切口，植入一个覆盖运动皮层大部分区域的传感器阵列——运动皮层是负责控制肢体运动的脑区。该手术耗时两个多小时，部分原因在于将传感器阵列连接至数据传输线缆的过程十分繁琐。不过，团队正在研发一种更为便捷的无线方案，或可将这一时间缩短至

40 分钟。

【Para.4】

The implants themselves allowed Dr Chang and his team to collect signals as the patients attempted to say particular words and make particular gestures. They then used those data to train two machine-learning models for each patient—one to decode the words and the other the gestures—allowing them to **reconstruct** both actions **concurrently** with approximately two-thirds accuracy. It is the first occasion that such **simultaneous** decoding has been achieved, allowing patients to communicate more **expressively**.

- **reconstruct v.** /ˌriːkənˈstrʌkt/ sth (from sth)to build or make sth again that has been damaged or that no longer exists 修复；重建；重造

They have tried to reconstruct the settlement as it would have been in Iron Age times.他们已试着按铁器时代的样子重建这个小村落。

- **concurrently adv.** /kənˈkʊrəntli/ 同时地，同时发生地，并发地；兼任地（兼）

The prison sentences will run concurrently. 所判的几个刑期合并执行。

- **simultaneous adj.** /ˌsɪmlˈteɪniəs/ happening or done at the same time as sth else 同时发生（或进行）的；同步的 【考研】

大纲词汇】

There were several simultaneous attacks by the rebels. 反叛者同时发动了几起攻击。

simultaneous translation/interpreting 同声传译

- **expressively adv.** /ɪk'spresɪvli/ with expression; in an expressive manner 富有表现力地，表情丰富地，意味深长地

He moved his hands expressively. 他很有表现力地移动了双手。

这些植入物使常博士及其团队能够在患者试图说出特定词语、做出特定手势时采集信号。随后，他们利用这些数据为每名患者训练两个机器学习模型——一个用于解码词语，另一个用于解码手势——从而能够以约三分之二的准确率同时重建这两种动作。这是首次实现此类同步解码，使患者能够更富有表现力地交流。

【Para.5】

There is further to go. These two initial individuals were restricted to a vocabulary of about ten words and phrases, such as “hello” and “nice to meet you”, and a similar number of gestures, such as a **clap** or a **shrug**. But Dr Chang and his collaborators are working on raising this to more than 100 actions. They also hope that the sensor array’s wide reach across the cortex means it may be possible to monitor not only intentions to communicate but also actions **intended** for the whole body. That might

permit paralysed individuals to operate devices such as robots and **exoskeletons**, which really would be life-changing.

- **clap** n. an act of clapping the hands; the sound this makes
鼓掌；拍手；掌声；拍手声 【考研大纲词汇】

Give him a clap! (= to praise sb at the end of a performance)
为他鼓掌吧！

- **shrug** n. /ʃrʌg/ to raise your shoulders and then drop them to show that you do not know or care about sth 耸肩（表示不知道或不在乎）【考研大纲词汇】

Sam shrugged and said nothing. 萨姆耸耸肩膀，什么也没说。

- **intend** v. to have a plan, result or purpose in your mind when you do sth 打算；计划；想要 【考研大纲词汇】

We finished later than we had intended. 我们完成时已超出原定时间。

- **exoskeleton** n. /'eksoskelɪt(ə)n/ (**biology** 生) a hard outer covering that protects the bodies of certain animals, such as insects (昆虫等动物的) 外骨骼

这项技术仍有很长的路要走。最初，这两名受试者可用的词汇仅限于约十个单词和短语，例如“你好”和“很高兴见到你”，以及同等数量的手势，例如拍手或耸肩。但常博士及其合作者正致力于将之提升至 100 多种动作。他们还希望，传感器阵列在大脑皮层上的广泛覆盖意味着，或许不仅能监测交流意图，还能监测针对全身的动作。这将使瘫痪者有望操作机器人、外骨骼等设备，从而真正改变其生活。

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➤ 今日翻译作业

(Para. 2) He and his colleagues have enabled two severely paralysed people to communicate with them, by means of an avatar, using both words and gestures.

（下一页提供句子分析）

➤ 今日句子分析

Para 2. He and his colleagues have enabled two severely paralysed people to communicate with them, by means of an avatar, using both words and gestures. 他和同事让两名严重瘫痪者能够借助虚拟形象, 运用言语和手势与研究人员交流。

主句

He and his colleagues have enabled two severely

主语

谓语

宾语

paralysed people to communicate with them, by means of

宾语补足语

方式状语

an avatar, using both words and gestures.

伴随状语

句子分析:

主句:

主语-He and his colleagues ; 谓语-have enabled; 宾语-two severely paralysed people; 宾语补足语, 说明宾语 “两名严重瘫痪者” 能够做什么- to communicate with them;; 方式状语- by means of an avatar, ; 伴随状语-using both words and gestures.

翻译解析：翻译难点不在生词，而在结构 enable sb. to do sth. 是复合宾语，整句结构是 enable people to communicate with them, by means of..., using..., 中文更自然的顺序为：主语+使+宾语+能够+借助方式+用手段+与对象+交流。

them 需要根据上下文，在中文字显化指代，以避免歧义，比如误以为是两位患者互相交流。

此外，by means of an avatar 与 using both words and gestures 两者都是方式状语，by means of an avatar 借助虚拟形象这一媒介，using both words and gestures：使用词语和手势这一手段，中文合并处理为：借助一个虚拟形象，用言语和手势.....” 。

➤ 写作积累

Para.1 Brain-computer interfaces have **come a long way** quickly.

come a long way 字面意思为“走了很远的路”，但 way 在很多场景下可以喻指“（事物）的发展/进展”，因此这个短语在此处引申为“（在某方面）取得很大进步、发展迅速、今非昔比”，例：

We've come a long way since the early days of the project.

这项目开始以来我们已取得很大进展。

与之相反的，have a long way to go 则指“还有很长一段路要走”→“还有很大的进步空间”，例：She still has a long way to go before she's fully fit. 她还需要很长时间才能完全恢复健康。

与 come a long way 相近的表达还有：

- ◆ **make significant strides** “取得大幅进展”：We're making great strides in the search for a cure. 在探索治疗办法方面，我们正不断取得重大进展。

- ♦ **advance by leaps and bounds** “突飞猛进” : Her health has improved in leaps and bounds. 她的健康已迅速好转。

Para.5 They also hope that the sensor array' s **wide reach** across the cortex means it may be possible to monitor not only intentions to communicate but also actions **intended for the whole body**.

reach 作名词时, 常表示“范围、触及面、影响力”, 例如以下几个常见表达:

- ♦ within reach: 在可及范围内, It saves time in the kitchen to have things you use a lot within reach. 在厨房里把常用的东西放在手边可以节省时间。
- ♦ beyond reach: 超出范围, The shot was well beyond the reach of the goalkeeper. 这次射门使守门员鞭长莫及。

文中这里的 wide reach 是指“物理/解剖学上的覆盖范围”, 指“传感器阵列铺在皮层上, 覆盖区域广”

而 actions intended for the whole body= 意图作用于整个身体的行动 / 针对全身的行动意图

其中 intended for 表示 “打算用于.....的 / 意图指向.....的 / 意图作用于.....的”，例：This book is intended for beginners.

这本书是给初学者用的。